

Q.22 Name the causative agent of enteric fever and mention one specific diagnostic test.

SECTION-D

Note: Long answer type questions. Attempt any two questions out of three questions. (2x8=16)

Q.23 Describe in detail five biochemical tests used in the identification of bacteria including their principle and result interpretation.

Q.24 Write in detail about the disc diffusion method for antibiotic sensitivity – include principle, procedure and precautions.

Q.25 Explain the laboratory diagnosis of respiratory tract, urinary tract and intestinal infections with sample collection and interpretation.

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Roll No.

1st Year / MLT

Subject : Clinical Micro-Biology

Time : 3 Hrs.

M.M. : 60

SECTION-A

Note: Multiple choice questions. All questions are compulsory (6x1=6)

Q.1 Which stain is used to detect Mycobacterium tuberculosis?

- a) Gram stain b) Acid-fast stain
- c) Methylene blue d) Negative stain

Q.2 Which of the following tests detects the presence of the citrate utilization?

- a) MR test b) Indole test
- c) Catalase test d) Citrate test

Q.3 A positive coagulase test confirms the presence of:

- a) E. Coli
- b) Staphylococcus aureus
- c) Streptococcus pyogenes
- d) Salmonella typhi

Q.4 Motility in bacteria is commonly observed using:

- a) Acid-fast staining
- b) Hanging drop method
- c) Simple staining
- d) Gram staining

Q.5 Which of the following organisms causes cholera?

- a) Shigella
- b) Proteus
- c) Vibrio cholera
- d) Streptococcus pneumoniae

Q.6 The disc diffusion method is used to assess:

- a) Glucose fermentation
- b) Bacterial motility
- c) Antibiotic sensitivity
- d) Indole production

SECTION-B

Note: Objective/ Completion type questions. All questions are compulsory. (6x1=6)

Q.7 The _____ method is used to determine bacterial motility

Q.8 A positive indole test produces a _____ ring after adding Kovac's reagent.

Q.9 _____ is the primary method for antibiotic susceptibility testing in microbiology labs.

Q.10 Treponema pallidum is a _____ shaped bacterium

Q.11 _____ test is used to detect the ability of bacteria to use citrate as a sole carbon source.

Q.12 Hospital acquired infections are called _____ infections.

SECTION-C

Note: Short answer type questions. Attempt any eight questions out of ten questions. (8x4=32)

Q.13 List any four general characteristics of bacteria.

Q.14 Define colony morphology and give two examples.

Q.15 Mention the principle of the MR-VP test.

Q.16 Name any two lactose fermenting and two non-lactose fermenting bacteria.

Q.17 Differentiate between Staphylococcus and streptococcus based on Gram reaction and arrangement.

Q.18 What is the importance of catalase and oxidase tests in bacterial identification?

Q.19 Define pathogenicity and give two examples of pathogenic bacteria.

Q.20 Name two common sources of nosocomial infections.

Q.21 Write a brief note on the laboratory diagnosis of wound infections.